

Bayesian Statistics: Techniques and Models

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Week 5 Capstone

Executive Summary

For my capstone, I picked the stars dataset from the dslab package. My objective with this data was to explore building a model that can predict the temperature of a star using the star's luminosity and the spectral class. Four different models were built and compared, linear, MCMC with a single feature, MCMC with a categorical feature, and MCMC with independent intercepts. The MCMC models with categorical feature and independent intercepts had better results when using the model to make predictions.

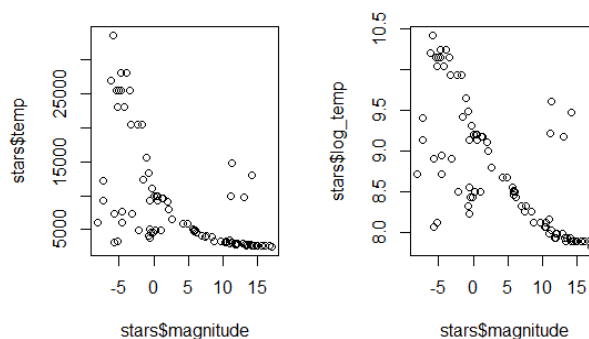
Introduction

The objective for this project is to find an interesting dataset, and use the methods learned from the course to gain insights on the dataset. I will do this by analyzing a selected dataset, create models for exploring the relationship between the features found in the dataset, and then evaluating the difference between the models.

Data

The data that I'm using for this project is the stars dataset that is part of the dslabs package. It contains data on the physical properties of selected stars, including luminosity, temperature, and spectral class. The features in the stars data are star (the unique name each individual star), magnitude (the luminosity and distance of the star), temp (the star's surface temperature in Kelvin), and type (a categorical spectral class of the star in the OBAFGKM system with ten categories).

I first plotted the relationship of the two numerical features, temp and magnitude. I see that there is a relationship between magnitude and temperature especially at the higher magnitude. The relationship looks exponential so I decided to take the log of temperature and plot that against magnitude. The log temperature plot shows a more linear relationship than the previous plot, so I'll continue to use the log temperature.

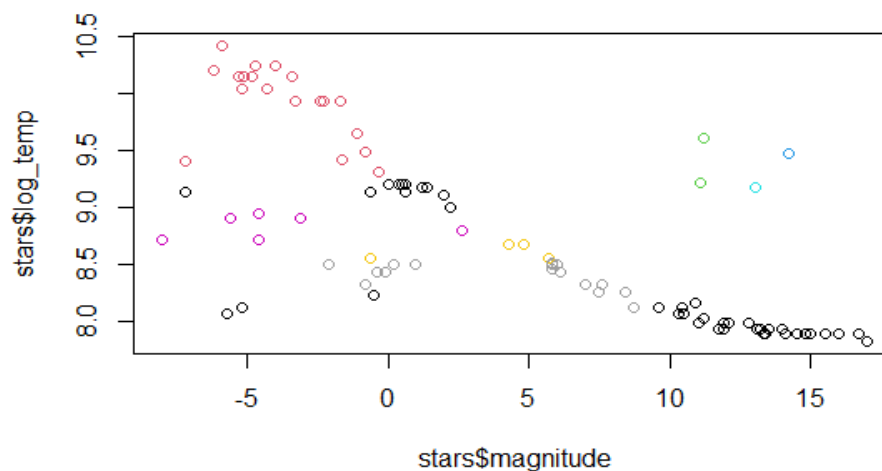


Models

Four models were used for exploring the relationship between magnitude and temperature for the stars dataset. The first model used was a simple linear model where $\log(temp) = b_0 + b_1 * magnitude$. In this model the magnitude is the independent variable and the log of the temperature is the dependent variable. This model was used as a baseline comparison to the MCMC based models that were also created.

The next model was implemented in JAGS with the same relationship between `log_temp` and `magnitude`. This model uses noninformative priors similar to what has been used in the lectures.

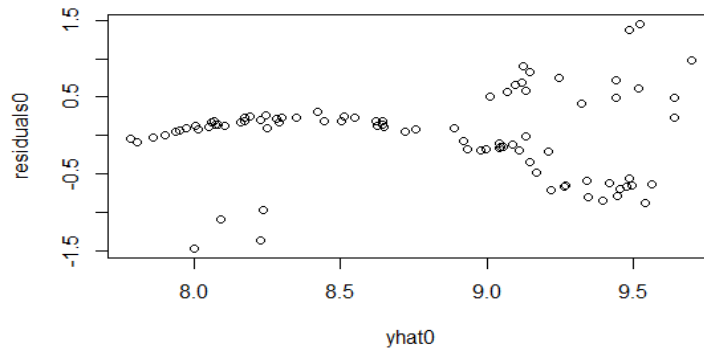
Next, I looked at adding an additional feature that specifies the spectral class of the star. This is stored as a categorical variable, so it'll need to be dummified before modeling. When plotting the magnitude and temperature you can see that there is some clustering of types in the plot.



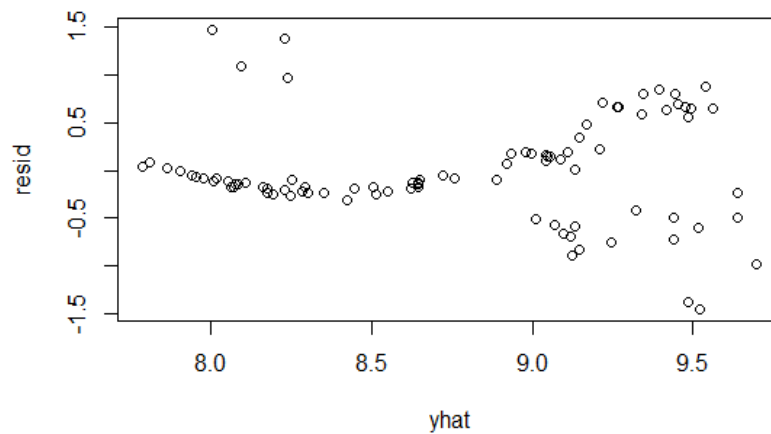
Finally, when looking at the plot with the star type we can see that there is some variance between groups. The last model that I tried is one where each star type has its own intercept.

Results

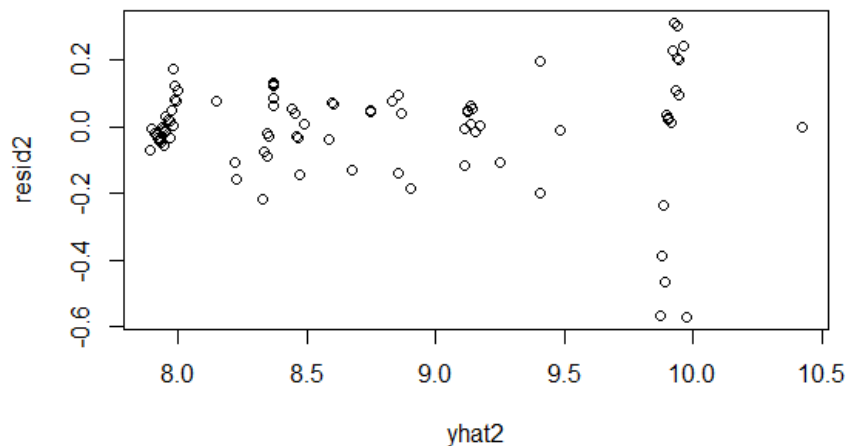
With the simple linear model, we can see that there isn't constant variance of the residuals which is one of the requirements for a linear model. Also, the RMSE from the linear model is 6079 which is very far from the actual values. I next looked at implementing a MCMC based models in order to improve on the RMSE.



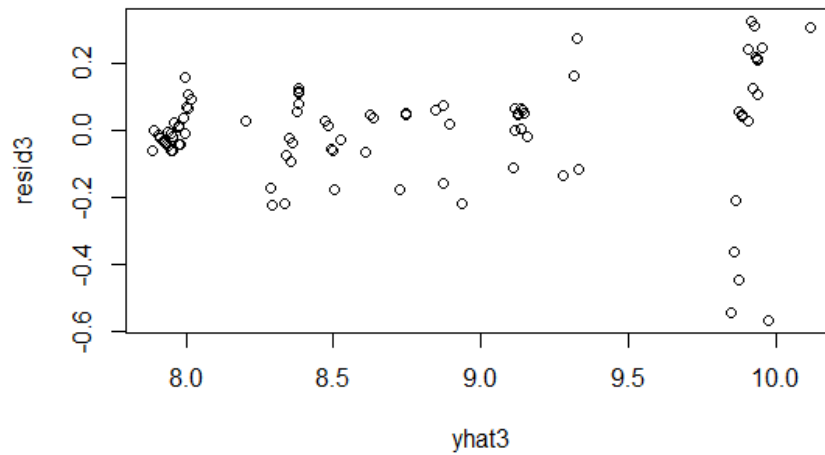
Looking at the plot for the residuals for the single feature MCMC model, we still see that the variance isn't constant. The penalized deviance for this model is 191.1 and the RMSE is 6081.396 which is like the value found with the simple linear model.



The variance of the residuals for the MCMC model with categorical features looks more evenly distributed than the previous models. The penalized deviance for this model is significantly less than the previous model (152 vs 191), and the RMSE is reduced as well (2431 vs 6082).



With the independent intercept model, the variance of the residuals is mostly constant except for the large values of $\hat{y}$. Also, this model has a lower penalized deviance (149) than the previous model, but the RMSE (2615) is slightly higher.



When comparing the different models using the RMSE and the DIC, we see that the MCMC based model with categorical features has the lowest RMSE, but the MCMC based model with independent intercepts has the lowest DIC.

##	model	RMSE	DIC
## 1	linear	6078.951	NA
## 2	MCMC Simple	6077.166	190.9
## 3	MCMC Cat Feature	2433.566	151.5
## 4	MCMC Indep Intercept	2615.479	149.0

Conclusion

The MCMC based models appear to give better predictions for the temperature of each star in the stars dataset given the magnitude than the linear model, and the best model to use would be either the model with the categorical model or the one with independent intercepts. Ultimately the decision would depend on whether deviance or RMSE is the more important metric. There is also some additional analysis that could be done, such as changing the prior distributions or parameters or increasing the number of iterations for the MCMC models.